Physics 1135 Version B

Fall 2025

Answer Sheet

EM Mini-Test

& Final Exam

Print LAST Name: _____

Rec Sec_____

First Name: _____

Remove this page from your exam when you begin. Write clearly in the space provided on this *Answer Sheet* the *letter* which you believe to be the *best* answer to each question.

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ONLY THIS ANSWER SHEET will be looked at for scoring. Make sure your chosen answers are on it and do not leave any answer space blank.

Neither calculators nor notes can be used during the test.

EM Mini-Test Score =

Final-Exam Score =

Mini-Test Responses (10 pts each)

Em - 1) D

Em - 2) C

Em - 3) B

Em - 4) A

Em - 5) A

Em - 6) C

Final-Exam Responses (10 pts each)

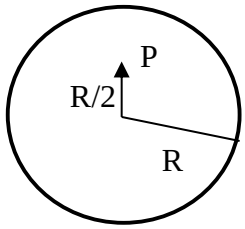
1) B
2) C
3) C
4) B
5) A
6) B
7) D
8) D
9) B
10) C
11) C
12) B
13) B
14) B
15) B
16) C
17) C
18) A
19) C
20) D

End-Material Mini Test

Em-1. A block attached to a spring is undergoing simple harmonic motion. The ratio of its kinetic energy to its total mechanical energy when it is two-thirds of the way to the maximum distance from its equilibrium position is:

A) 2/3 B) 1/9 C) 1/3 D) 5/9

Em-2. A circular disk of radius R and mass M has moment of inertia ½MR^2 about an axis through its center and perpendicular to its surface. The disk is pivoted at point P distance ½R from the center. The disk's period for small oscillations is:



A) $\sqrt{\frac{2g}{3R}}$ B) $2\pi\sqrt{\frac{R}{g}}$ C) $2\pi\sqrt{\frac{3R}{2g}}$ D) $2\pi\sqrt{\frac{2R}{g}}$

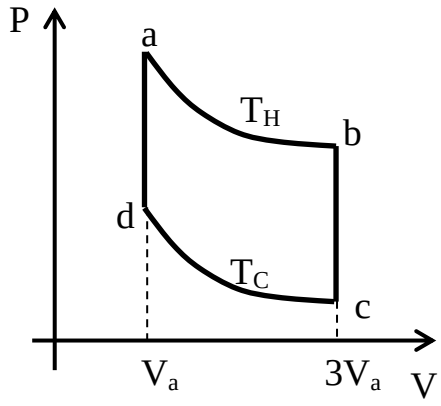
Em-3. A string of length 60 cm has fixed ends. Two consecutive harmonics are 400Hz and 300 Hz. The wave speed equals:

A) 180 m/s B) 120 m/s C) 60 m/s D) 12 m/s

Em-4. A dolphin is swimming directly toward a vertical seawall at one-fourth the speed of sound in water. It is generating sound waves in the water of frequency f. The frequency of the echo that the dolphin hears is:

A) ($^5/_3$) f B) f C) ($^3/_2$) f D) ($^{16}/_9$)f

Questions Em-5 and Em-6 both refer to the cycle in the figure on the left.



Em-5. A **diatomic** ideal gas goes through cycle a→b→c→d→a as shown in the figure. Processes a→b and c→d are isothermal and occur at temperatures T_H and T_C, respectively. The total work done by the gas in the cycle is:

A) nR (T_H-T_C) ln 3 B) nR(T_C-T_H) ln 3

C) $^3/_2$ nR (T_H-T_C) D) $^5/_2$ nR (Tc-T_H) ln 3

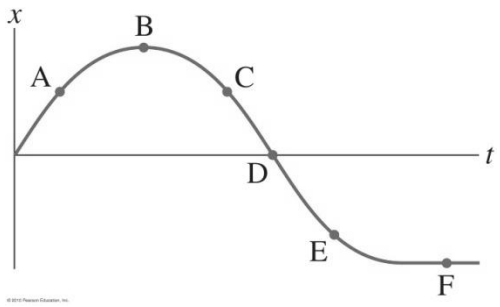
Em-6. For the cycle in the figure, the heat flowing into this diatomic gas during the process d→a equals:

A) zero B) nR (T_H-T_C) C) $^5/_2$ nR (T_H-T_C) D) $^5/_2$ nR($2V_a$)

Final Exam

Ignore air resistance for all problems.

1. The motion of an object is described by the *x*-*t*-diagram at the right. Which of the following is true?

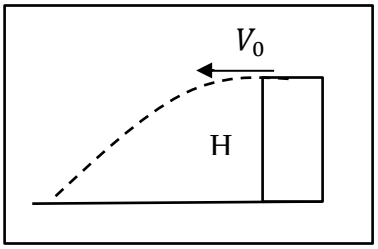


A) The object moves in the negative x-direction at A.
B) The object stops at B.
C) The object is faster at F than at C.
D) The object stops at D.

2. The position of an object in one dimensional motion as a function of time t is given by $y(t) = at^3 + bt^2 - c$ where a, b, and c are positive constants. At time $t = 0$, the object's position, velocity component, and acceleration component are:

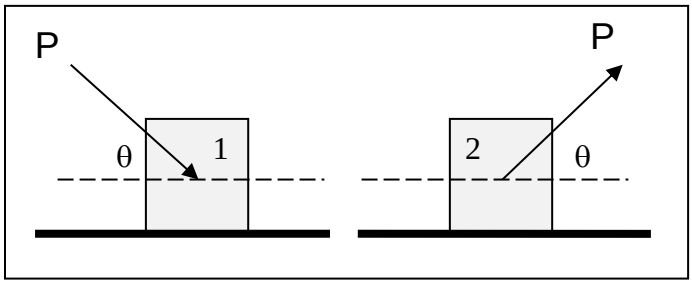
A) $y = -c$, $v_y = 0$, and $a_y = 0$
B) $y = 0$, $v_y = 2b$, and $a_y = 0$
C) $y = -c$, $v_y = 0$, and $a_y = 2b$
D) $y = -c$, $v_y = 2b$, and $a_y = 0$

3. A ball is kicked from a vertical height *H* with initial horizontal velocity V_0. What horizontal distance will the ball travel before hitting the ground below?



A) $V_0\sqrt{\frac{g}{2H}}$ B) $\frac{HV_0}{g}$ C) $V_0\sqrt{\frac{2H}{g}}$ D) $\frac{2HV_0}{g}$

4. Two identical blocks are on the same rough horizontal surface. Block 1 is **pushed** with a force of magnitude P at angle θ with the horizontal. Block 2 is **pulled** with a force of magnitude P at angle θ above the horizontal. The magnitude of kinetic friction



A) is smaller on block 1 than on block 2.
B) is larger on block 1 than on block 2.
C) is the same for both blocks.
D) depends on the acceleration.

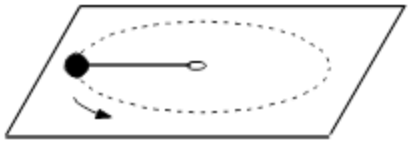
5. An object with mass *m* is on a scale in an elevator which is moving down at a constant acceleration *a*. What is the reading on the scale?

A) *mg – ma* B) *mg + ma* C) *mg* D) *ma*

6. A particle moves with **constant speed** along a curved path that is never straight. The net force acting on the particle is:

A) zero
B) always perpendicular to the particle's velocity
C) constant
D) always parallel to the particle's velocity

7. A flat puck is moving in a circle on a horizontal frictionless table. It is held in its orbit by a string. The puck takes 2 seconds to complete a full circle. The tension in the string equals 8N. If the puck went around the same circle in one second, what would be the tension in the string?



A) 2N B) 4N C) 24N D) 32N

8. The potential energy of a force is given as $U(x)=ax^2 + bx^3+c$. The x-component of the force is

A) $\frac{1}{3}\, ax^3 + \frac{1}{4}\, bx^4$ B) $-\frac{1}{3}\, ax^3 - \frac{1}{4}\, bx^4$ C) $2\, ax + 3\, bx^2$ D) $-2\, ax - 3\, bx^2$

9. A block of mass M sits on a vertical spring of force constant k that is compressed a distance S from its equilibrium length. The block is then released from rest and shoots up, leaving the spring behind. The speed of the block when it reaches height H above the starting position is

A) $\sqrt{2gH}$ B) $\sqrt{\frac{kS^2}{M} - 2gH}$ C) $\sqrt{\frac{kS^2}{2Mg}}$ D) $\sqrt{\frac{kS^2}{M} + 2gH}$

10. An incline makes an angle θ with the horizontal. A constant pushing force F that is directed **parallel to the incline** pushes a block of mass M up the incline by a distance L along the incline. The work done by the pushing force equals

A) $FL \cos\theta$ B) $FL \sin\theta$ C) FL D) $(F - Mg \sin\theta)\, L$

11. A conservative force acts on a particle. We can conclude:

A) The force must be constant.

B) The acceleration of the particle will be zero.

C) The work done by the force on the particle depends only on initial and final position.

D) The particle cannot change its direction.

12. Planet A and planet B have the same radii. Planet A has an escape speed that is twice the escape speed from Planet B. If planet B has mass M, planet A has mass

A) $16\, M$ B) $4\, M$ C) $2\, M$ D) $\frac{1}{4}\, M$

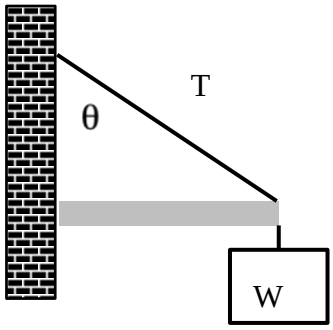
13. A ball of mass m made of clay moves in the positive x-direction with velocity v. Another ball, of the same mass, coming up from behind with velocity $3v$ in the positive x-direction, is colliding with the first ball. The balls stick together after the collision. What is the speed of the combined object?

A) v B) $2v$ C) $3v$ D) $4v$

14. A boy of mass m stands at the center of a merry-go-round of mass M and radius R that is initially rotating with angular speed ω about an axle through the center. The moment of inertia of the merry-go-round is $\frac{1}{2}\, MR^2$. The boy walks from the center to the edge of the merry-go-round. Consider the boy to be a point mass. The angular speed of the merry-go-round when he reaches the edge is:

A) $\frac{M+2m}{2M}\omega$ B) $\frac{M}{M+2m}\omega$ C) $\frac{M}{2M+m}\omega$ D) $\frac{M+2m}{M}\omega$

15. A beam of weight 2W is attached to a wall and held horizontal by a massless cable which makes an angle θ with respect to the wall. There is a weight W hanging from the end of the beam. What is the tension in the cable?



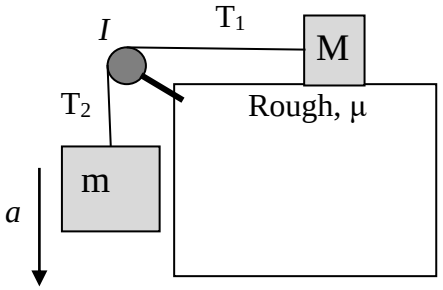
A) $\frac{3W}{sin\,\theta}$ B) $\frac{2W}{cos\,\theta}$ C) $\frac{2WL}{sin\,\theta}$ D) $\frac{2W}{sin\,\theta}$

16. A student spins on a rotating chair, holding two weights close to his body. When the student stretches out his arms, the angular speed
A) increases because the moment of inertia decreases.
B) increases because the moment of inertia must remain the same.
C) decreases because the moment of inertia increases.
D) decreases because the moment of inertia must remain the same.

17. A solid uniform ball of mass *M* and radius *R* is on an incline a height *H* above the base of the incline. The moment of inertia for a uniform ball about its center of mass is $I = \frac{2}{5}MR^2$. The angle of the incline with the horizontal is θ. After the ball is released from rest, it rolls without slipping down the incline. What will the speed of the ball be when it reaches the bottom of the incline?

A) $\sqrt{2gH}$ B) $\sqrt{5gH/2}$ C) $\sqrt{10gH/7}$ D) $\sqrt{4gH/3}$

18. Two blocks of mass *M* and *m*, respectively, are connected by a string, which runs over a pulley of moment of inertia *I*. The pulley's axle is frictionless, and the string does not slip on the pulley. The coefficient of kinetic friction on the horizontal surface is such that the hanging block of mass *m* accelerates downwards. In that case, the tensions T_1 and T_2 in the strings must satisfy:



A) $T_1 < T_2$ B) $T_1 = T_2$ C) $T_1 > T_2$ D) $T_1 < 0$

19. A uniform disk of mass M, radius R and moment of inertia $I = \frac{1}{2}MR^2$ is rolling without slipping down a slope that is inclined by an angle θ with respect to the horizontal. The disk's linear acceleration equals

A) $(2/3)sin\theta$ B) $(3/2)gsin\theta$ C) $(2/3)gsin\theta$ D) $(2/3)cos\theta$

20. An object is in static equilibrium if, and only if,
A) the sum of the external forces is zero.
B) all forces that are acting on the object are conservative.
C) there are no forces acting on the object.
D) the sum of the external forces and the sum of the external torques are both zero.

The Physics 1135 instructors wish you a wonderful winter break!